Introduction to the Turkish Economy

Lecture 1

Learning objectives

- **Interpret GDP, growth, inflation and labor-market indicators.**
  GDP measures production in a period, and real GDP growth tracks changes in output after allowing for prices. Inflation and unemployment are assessed separately because they affect purchasing power and job opportunities, respectively.
- **Describe the sectoral structure and external links of Türkiye's economy.**
  Sectoral value added shows what the economy produces, and sectoral employment shows where people work. The trade and energy sections explain why imports matter for domestic production and investment.

Learning objectives

- **Explain why average income does not fully measure living standards.**
  Per capita income gives an average across the population and cannot show who receives the income. Income distribution and indicators such as education and health give a fuller picture of living conditions.
- **Connect economic evidence to choices in public administration.**
  Inflation, employment and poverty affect household welfare and help identify priorities for public policy. Applying the evidence means identifying which people are affected and selecting an indicator that matches the policy objective.

Measurement of economic performance

- **If total output rises, is everyone better off?**
  Higher GDP means more production, but its benefits can be distributed unevenly. Output and per capita income capture only some aspects of economic development.
- **Think about prices, jobs and the distribution of income.**
  Price increases can reduce what a fixed income buys, even while total production grows. Unemployment and the division of income among households reveal effects that the aggregate output figure misses.
- **Give one example in which GDP rises but a household becomes worse off.**
  Suppose production rises in export industries while one worker loses a job. If the household also pays higher prices for essentials, it can afford less even as national output increases.

Measurement of economic performance

Question	What to check
What is measured?	Output, prices, employment, income or a stock of assets
In which units?	TRY, US dollars, PPP dollars, percentages or people
Over which period?	Year, quarter, month, average or end-of-period
Compared with what?	Earlier years, other countries or a policy target
What does it establish?	A description, an association or a causal explanation

- **What is measured?**
  GDP measures production, while the CPI measures the cost of a consumer basket and unemployment describes people without work in the labor force. Each measure answers a different question about the economy.
- **In which units?**
  GDP in current dollars is sensitive to the exchange rate, while real GDP in lira is used to follow domestic output growth. A labor-market count in people becomes a rate only after choosing a denominator.

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- **Over which period?**
  GDP refers to production over an interval, such as a quarter or a year. CPI changes can be measured month to month, against the same month a year earlier, or by comparing twelve-month averages.
- **Compared with what?**
  Income can be compared across countries and over time using consistent price measures. Comparing realized growth with development plan targets shows how outcomes differed from stated objectives.

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- **What does it establish?**
  The relationship between inflation and growth differs across the countries examined. A reported correlation describes how the variables move together and does not determine which causes the other.

Gross domestic product

- **GDP measures the value of final goods and services produced within an economy during a period.**
  GDP is measured at market prices and includes final goods and services. The total covers a flow of production over a period such as a quarter or a year.
- **"Domestic" refers to production within the territory.**
  GDP follows the location of the activity, so output generated within the country enters domestic production. GNP, by contrast, adjusts domestic production for income received from and paid to the rest of the world.

Gross domestic product

- **"Final" prevents counting intermediate inputs twice.**
  When calculating production by value added, subtract the cost of materials and services used as inputs from the value of output. Adding the resulting value added across producers avoids counting the same input again in the final product.
- **"Gross" means that consumption of fixed capital has not been deducted.**
  A net measure deducts the value of capital used up in production. This matters because part of gross production compensates for the use of machinery, buildings and other fixed assets.

Currency conversion and purchasing power

Measure	Useful for	Main caution
Real GDP in domestic currency	Output growth over time	Keep the volume methodology consistent
GDP in current US dollars	Economy size at market exchange rates	Exchange-rate changes affect the value
GDP per capita at PPP	Comparing purchasing power across countries	An average conceals income differences

- **Real GDP in domestic currency**
  Real output measures remove the effect of price changes so that growth can be compared over time. Türkiye's real GDP series moved from fixed base-year prices to chain-linked volume measures.
- **GDP in current US dollars**
  A market exchange rate translates the current lira value of production into US dollars. A change in the lira's dollar value can therefore move dollar GDP even when the change in domestic output is smaller.

Currency conversion and purchasing power

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GDP in current US dollars	Economy size at market exchange rates	Exchange-rate changes affect the value
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- **GDP per capita at PPP**
  "Purchasing power parities, abbreviated as PPPs, are indicators of price level differences across countries." Converting per capita income with PPP helps compare purchasing power, while the resulting figure is still an average across people.

GDP per capita and living standards

- **GDP per capita** $= \dfrac{\textbf{GDP}}{\textbf{population}}$

  GDP per capita assigns an equal arithmetic share of total output to each person. It is a measure of average output, and a household's actual income can be very different.

- **A larger economy need not have higher average income.**

  Total GDP measures the size of an economy, while GDP per capita measures output per person. Dividing by population can change the ranking because countries with similar total output can have very different population sizes.

GDP per capita and living standards

- **Real income per person rises when output grows faster than population.**
  Real GDP per capita divides the volume of production by the number of people. If real output increases faster than population, that ratio increases even though some households may lose income.
- **Health, education, distribution and environmental quality also matter.**
  Per capita income alone does not describe income distribution or living conditions. It can be supplemented with indicators of health, education, and social and environmental conditions.

GDP by the expenditure approach

- $Y = C + I + G + X - M$
  The expenditure approach adds purchases of final goods and services by households, firms, governments and foreign buyers, with an adjustment for imports. It measures the same production as the value-added and income approaches in principle.
- **C: household consumption. I: investment in fixed capital and inventories.**
  Consumption includes household purchases such as food, appliances and transport services. Investment includes new buildings and equipment, additions to inventories and purchases of newly built dwellings.
- **G: government purchases of goods and services.**
  Public purchases include the resources used to provide services, such as staff services and road works. Transfers such as pensions do not themselves buy current production for the government.

GDP by the expenditure approach

- **X − M: exports minus imports of goods and services.**
  Exports are domestically produced goods and services bought by foreign residents. Imports are goods and services produced abroad and bought by residents, so the difference is net exports.
- **Imports are subtracted to remove foreign production already included in spending.**
  Some purchases in consumption, investment or government spending are produced abroad. Subtracting imports removes that foreign production from the domestic expenditure total.

GDP by the expenditure approach

Transaction	Classification
A household buys locally produced bread	C
A firm buys a newly produced machine	I
A public institution buys a service	G
A household buys an imported refrigerator	C and M
The government pays a pension	Transfer, not a direct GDP purchase
An investor buys an existing share	Financial transaction, not I in GDP

- **A household buys locally produced bread** Bread purchased for household use is consumption of a final good. The purchase enters C, and its domestic production contributes to GDP.
- **A firm buys a newly produced machine** The machine adds to the firm's stock of physical capital. The expenditure therefore enters I, like other purchases of newly produced machinery and equipment.

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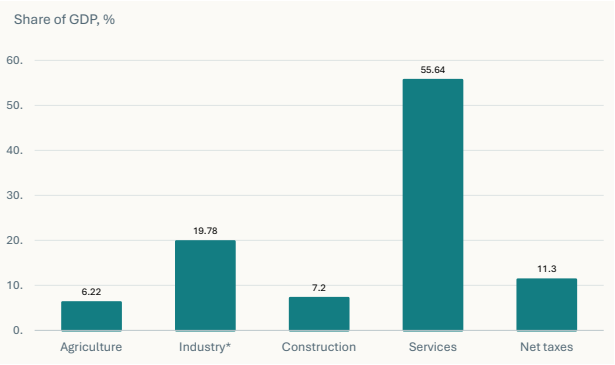
- **A public institution buys a service**
  The institution receives a service in exchange for the payment. This is government expenditure on current production and enters G.
- **A household buys an imported refrigerator**
  The refrigerator is a durable good and its purchase is recorded in C. The import entry removes the value of foreign production from domestic GDP.

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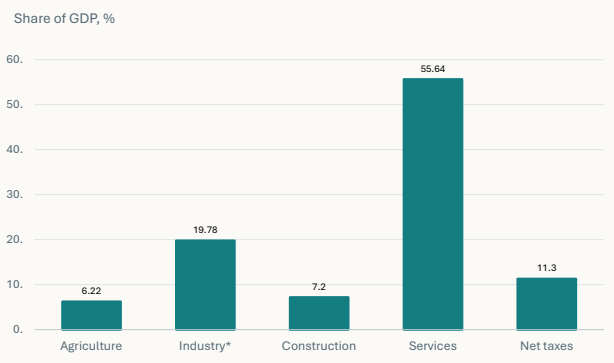
- **The government pays a pension**
  The pension is a transfer because the government receives no current good or service in return. A recipient's later purchase of goods and services can enter consumption.
- **An investor buys an existing share**
  Buying an existing share changes ownership of a financial asset. It is excluded from investment in GDP, which covers newly produced physical capital and changes in inventories.

Sectoral composition of GDP



- **Services account for more than half of reported GDP.**
  "Since 1998, more than half of the total value-added or production has belonged to the services sector in Turkey." The aggregate includes activities as different as retail, transport, finance, education and public administration.
- **Industry and construction support demand for inputs and labor.**
  A sector affects other activities through the inputs it buys and the goods or services it supplies. Industry here includes mining, manufacturing and utilities, with construction shown separately.

Sectoral composition of GDP



- **A sector's size does not show its productivity or technology level.**
  The share measures the sector's contribution to GDP. Value added per worker provides an approximate measure of labor productivity.

Sectoral composition of GDP

Share of GDP (%)	1998	2008	2018
Agriculture	9.93	7.44	6.22
Industry*	19.33	19.22	19.78
Construction	4.66	6.37	7.20
Services	56.63	55.50	55.64
Taxes less subsidies	10.66	11.25	11.30

* Industry includes mining, manufacturing and utilities.

- **Agriculture**
  Agriculture's share falls from 9.93% in 1998 to 6.22% in 2018. Its declining share means other activities grew in relative importance, without establishing that farm output fell in absolute terms.
- **Industry***
  The broad industry share stays near one fifth in all three years shown. Industry includes mining, manufacturing, electricity and water-related activities in this series.
- **Construction**
  Construction rises from 4.66% to 7.20% in the selected years. Housing and construction were important areas of growth during the 2000s.

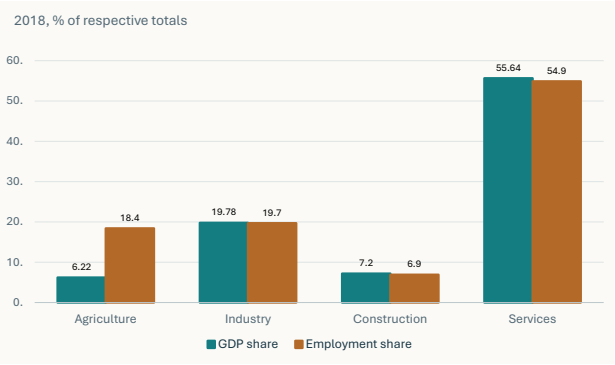
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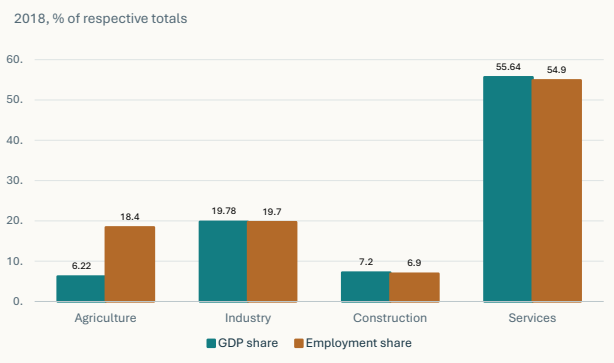
- **Services**
  Services remain the largest group, with shares around 55–57% in these observations. The aggregate covers both market services and activities such as education, health and public administration.
- **Taxes less subsidies**
  GDP at purchaser prices includes an adjustment for product taxes less subsidies. The row is therefore an accounting adjustment rather than an additional producing sector.

Output shares and employment shares



- **Agriculture employs 18.4% of workers but produces 6.22% of reported GDP.**
  The GDP share and employment share have different denominators. The 2018 comparison shows that agriculture's share of workers was much larger than its share of measured output.
- **The gap points to lower average output per worker.**
  A sector's output share divided by its employment share gives a rough comparison with output per worker for the economy. Earlier estimates of value added per worker also place agriculture below industry and services.

Output shares and employment shares



- **Hours, capital, informality and measurement also affect the comparison.** Informal activity can affect how much output and employment are recorded. Value added per worker is a rough average because it does not isolate labor effort or adjust for the capital used.

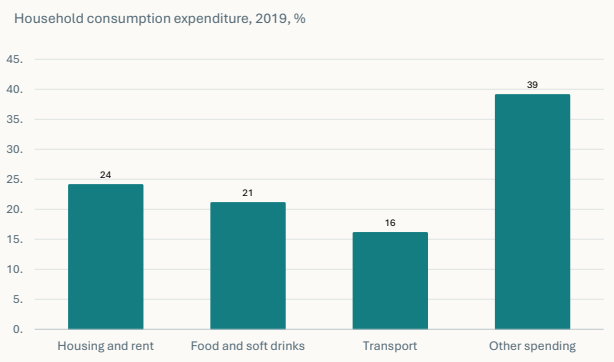
Structural change and public policy

- **Moving workers to more productive activities can raise average income.**
  Employment and production shifted away from agriculture, and value added per worker differed across sectors. When workers can find higher-productivity jobs, a change in the employment mix can raise aggregate output per worker.
- **Skills, capital and infrastructure affect which jobs become available.**
  Investment expands productive capacity, while human development and technological change can support long-run growth. Applying these ideas, the jobs created depend on both worker skills and the complementary facilities firms can use.

Structural change and public policy

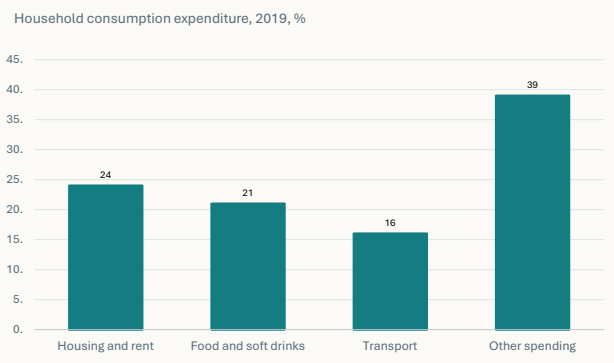
- **Housing and transport influence whether workers can move.**
  Housing and transport are two of the largest categories of household spending. As a classroom application, those costs can determine whether a worker can afford to relocate or commute to a new job.
- **Agricultural productivity and rural public services remain important.**
  In the sectoral comparison, value added per worker was much lower in agriculture than in industry or services. Improving rural services and productive capacity is a possible response for people who continue to work in agriculture.

Composition of household expenditure



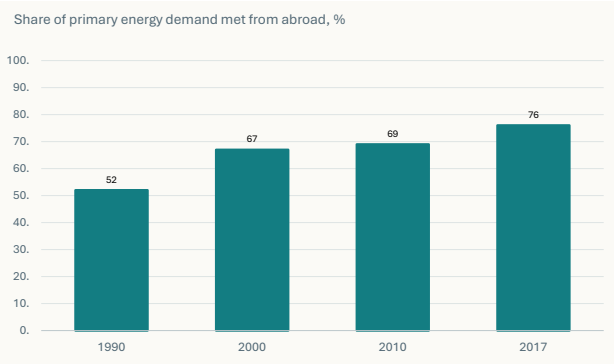
- **These first three categories account for 61% of the budget.**
  In 2019, housing and rent accounted for 24%, food and non-alcoholic drinks for 21%, and transport for 16%. Together they made up 61% of measured household consumption spending.
- **Price increases in essentials can sharply affect living standards.**
  The CPI tracks the cost of a basket, so price changes in large spending categories can matter considerably. A household whose income fails to keep pace can buy less of that basket.

Composition of household expenditure



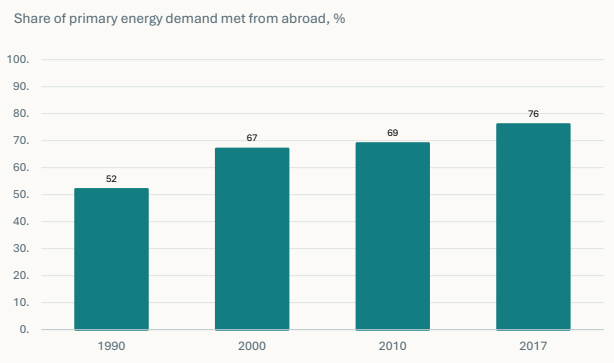
- **The national average does not describe every household.**
  The chart summarizes aggregate consumption shares and does not report separate baskets for income groups. Because income is unevenly distributed, the same price change can affect households differently depending on what they buy.

Dependence on imported energy



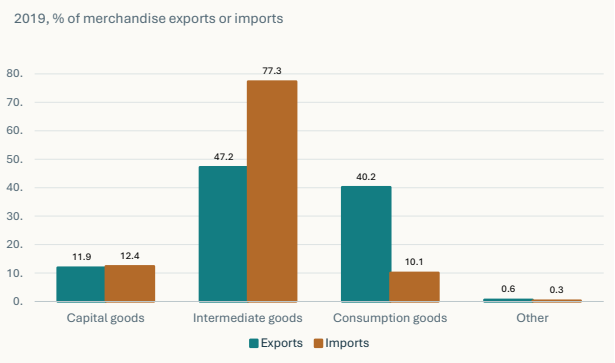
- **The imported share of primary energy demand rose from 52% to about 76%.** The share of primary energy demand met from abroad rose from 52% in 1990 to around 76% in 2017. At the latter date, domestic sources met roughly one quarter of primary demand.
- **Energy prices can affect import spending and production costs.** Türkiye relies heavily on imported oil and gas, and industry and transport are major energy users. A foreign price increase can therefore raise the import bill and the costs of energy-using firms.

Dependence on imported energy



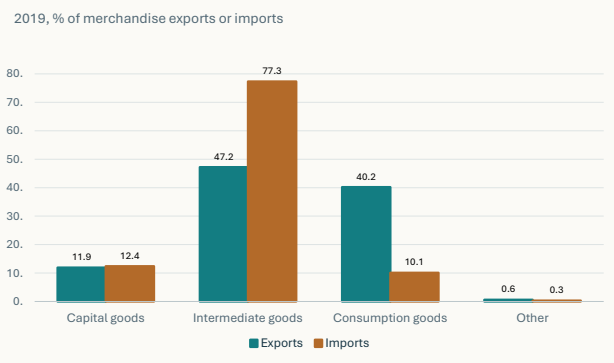
- **Efficiency and domestic supply can change this exposure over time.**
  The dependence ratio compares foreign supply with total primary energy needs. Lower energy use for a given level of output, or more domestic supply replacing imports, would reduce exposure, other things equal.

The composition of foreign trade



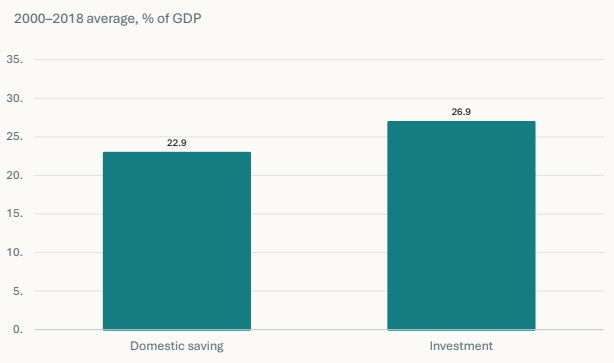
- **Intermediate and capital goods make up about 90% of imports.**
  "Approximately 90% of Turkish imports consist of intermediate and capital goods." The 2019 shares are 77.3% and 12.4%, respectively.
- **Imports can support domestic production and investment.**
  Intermediate imports supply materials and components used in current production. Capital imports provide machinery and other equipment, so a high import share can also indicate inputs into domestic activity.

The composition of foreign trade



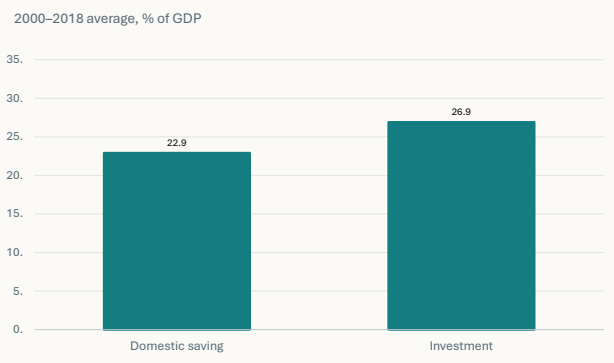
- **Import composition does not equal the foreign content of exports.**
  The import shares classify everything imported, while foreign content measures imported value embedded in exports. A separate estimate puts the imported content of Türkiye's exports at 16.5%.

Saving and investment



- **Average investment exceeds domestic saving by 4 percentage points.** From 2000 to 2018, total investment averaged 26.9% of GDP and gross national saving averaged 22.9%. The difference was 4 percentage points of GDP.
- **External finance helps cover the difference.** When investment exceeds national saving, resources from abroad cover the aggregate gap. Earlier national accounts also show a domestic saving shortfall.

Saving and investment



- **The terms and use of that finance matter for sustainability.**
  Persistent reliance on external financing can increase vulnerability to currency crises. For evaluation, ask whether the resources build productive capacity and whether the financing can be sustained if external conditions tighten.

The current account and the saving gap

- **Current account = national saving − investment**
  With national saving measured from national disposable income, the balance equals national saving minus investment. A simplified goods-market identity uses net exports and separates the private saving gap from the government balance.
- **A deficit corresponds to investment exceeding saving.**
  A negative saving-minus-investment balance indicates that national saving falls short of investment. Part of Türkiye's investment was financed by foreign saving during the period examined.

The current account and the saving gap

- **Borrowing and equity flows can finance the gap.**
  Foreign saving provides financing when domestic resources fall short. Borrowing creates a repayment obligation, while equity gives foreign investors an ownership claim and a share of future returns.
- **A smaller deficit during a recession may reflect falling investment and imports.**
  Türkiye's saving gap narrowed in the 2001 and 2009 recessions. A contraction can reduce investment demand and imports, so a smaller gap alone does not show improved productive capacity.

Economic growth: level and rate

- **Growth (%)** $= \left(\frac{Y_t}{Y_{t-1}} - 1 \right) \times 100$

  Growth is calculated as the change from the previous period divided by the previous period's level. The expression on the slide is the same calculation written as a ratio minus one.

- **Use real GDP to measure output growth.**

  A rise in current-price GDP can reflect higher prices even with unchanged quantities. Real GDP is therefore used to measure output growth.

Economic growth: level and rate

- **Use real GDP per capita to measure growth in average output per person.**
  Dividing real GDP by population adjusts for changes in the number of people sharing total output. Real GDP per capita can be used to compare long-run income trends across countries.
- **A long-run average can conceal recessions and uneven progress.**
  The long-run growth rate summarises the change between an initial and a final observation. The annual series shows sharp declines around crises even when the longer-run trend is upward.

Long-run growth in comparative perspective

Country	1923 income	2016 income	Annual growth
South Korea	871	36,103	4.1%
Türkiye	1,242	17,906	2.9%
Japan	2,847	37,465	2.8%
Germany	5,554	44,689	2.3%
United States	9,420	53,015	1.9%

Real GDP per capita in 2011 US dollars.

- **South Korea**
  South Korea's real GDP per capita rose from 871 to 36,103 in 2011 US dollars. Its 4.1% annual growth rate is the highest among the five selected rows.
- **Türkiye**
  Real GDP per capita rises from 1,242 to 17,906, implying about 2.9% average annual growth. The starting level was above South Korea's, but the 2016 level was much lower.
- **Japan**
  Japan starts at 2,847 and reaches 37,465 in the reported series. Its growth rate is close to Türkiye's, but its higher starting level helps explain its higher final level.

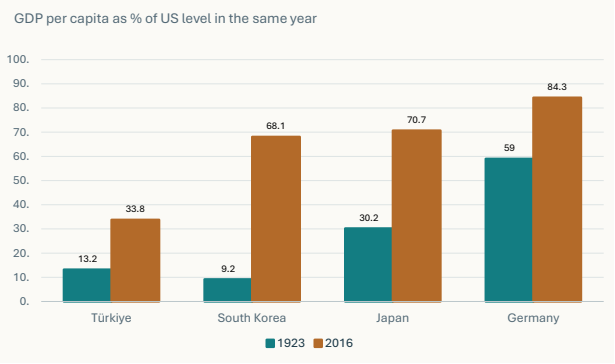
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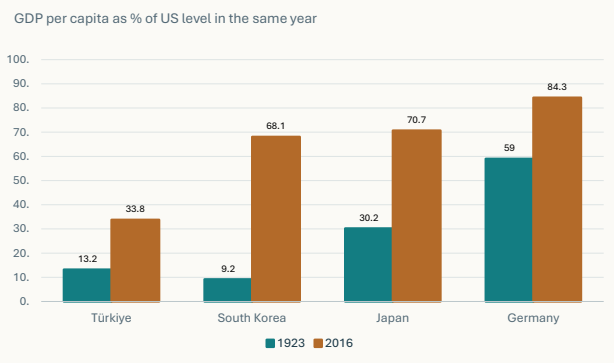
- **Germany**
  Germany's per capita income grows from 5,554 to 44,689 in the comparison. The 2.3% rate is lower than Türkiye's, while the level stays higher at both endpoints.
- **United States**
  The United States has the highest initial and final income among these rows despite the lowest average growth rate. The contrast shows why comparing income levels and comparing growth rates are different exercises.

International income convergence



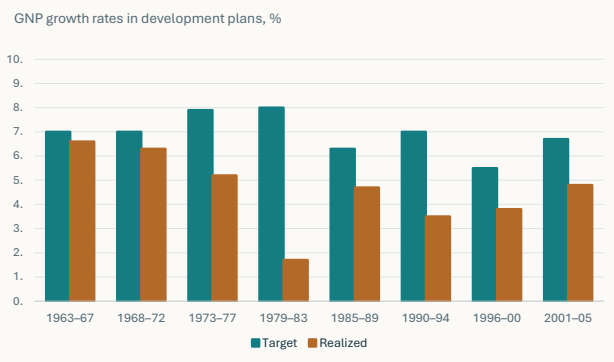
- **Türkiye rises from about 13% to 34% of the US level.**
  The ratios divide Türkiye's real GDP per capita by the US figure in each year. Türkiye catches up in relative terms while remaining below the US level at the second endpoint.
- **South Korea closes a larger part of the gap.**
  South Korea rises from 9.2% to 68.1% of US real GDP per capita. It starts below Türkiye on this measure and ends at around twice Türkiye's relative level.

International income convergence



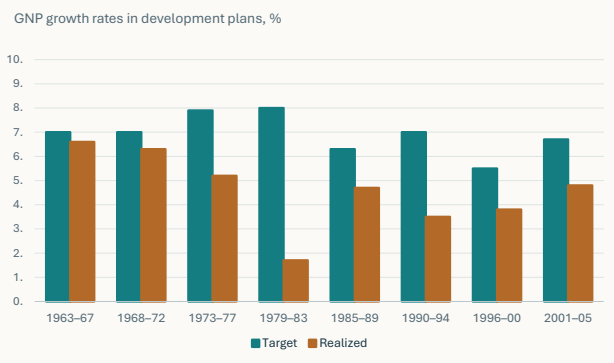
- **The pattern raises questions about productivity, skills and institutions.** Technology, human development and investment can help explain differences in long-run growth. The two endpoint ratios establish the difference in outcomes and provide a starting point for investigating its causes.

Growth targets and outcomes in development plans



- **Realized growth falls below the target in each listed plan.**
  The chart compares targeted and realized GNP growth across development plans. For example, the First Plan targeted 7.0% and realized 6.6%, while the Fourth targeted 8.0% and realized 1.7%.
- **Targets express intentions. Outcomes also reflect shocks and implementation.**
  The plans assigned growth targets to guide development, especially in the earlier planning period. Economic and political crises contributed to the unusually large shortfall under the Fourth Plan.

Growth targets and outcomes in development plans



- **The gaps motivate policy evaluation.** Growth targeting became less central as economic decisions shifted toward markets after 1980. The table also notes changes in national-account bases, so compare the periods with attention to both policy and measurement.

Alternative approaches to growth

Approach	Main emphasis	Question for evaluation
Import-substituting industrialization	Domestic productive capacity and planning	Does protection support learning and productivity?
Export-led growth	Foreign markets and private investment	Can firms compete and expand domestic value added?
Knowledge and capabilities	Skills, innovation and technology	Can productivity gains persist?

- **Import-substituting industrialization**
  In the planned period, policy aimed to build domestic industry and replace selected imports with domestic production. The Second Plan put more emphasis on producing intermediate and investment goods at home.
- **Export-led growth**
  Export promotion was central to the growth strategy adopted after 1980. In this account, private investment decisions and conditions in foreign markets matter more for outcomes.

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- **Knowledge and capabilities**
  Research and development and information and communication technologies can drive technological change within the economy. Sustained productivity growth then depends on the capacity to develop and apply knowledge.

Labor-market concepts

Concept	Definition
Employed (E)	People classified as employed during the reference period
Unemployed (U)	People without work who seek and are available for work
Labor force (L)	Employed plus unemployed: $L = E + U$
Participation rate	Labor force $\div$ relevant working-age population
Unemployment rate	Unemployed $\div$ labor force
Employment rate	Employed $\div$ relevant working-age population

- **Employed (E)**
  People who worked during the reference period as employees, employers, self-employed workers or unpaid family workers are counted as employed. The reported group is broader than people with regular salaried jobs.
- **Unemployed (U)**
  To be counted as unemployed, a person must be without employment, be looking for work and be available to start. A person who is without a job but has stopped searching can fall outside the labor force.

Labor-market concepts

Concept	Definition
Employed (E)	People classified as employed during the reference period
Unemployed (U)	People without work who seek and are available for work
Labor force (L)	Employed plus unemployed: $L = E + U$
Participation rate	Labor force $\div$ relevant working-age population
Unemployment rate	Unemployed $\div$ labor force
Employment rate	Employed $\div$ relevant working-age population

- **Labor force (L)**
  The labor force adds employed and unemployed people. The working-age population also includes people outside the labor force.
- **Participation rate**
  Participation divides the labor force by the working-age population. It can rise when previously inactive people start working or looking for work.

Labor-market concepts

Concept	Definition
Employed (E)	People classified as employed during the reference period
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Participation rate	Labor force $\div$ relevant working-age population
Unemployment rate	Unemployed $\div$ labor force
Employment rate	Employed $\div$ relevant working-age population

- **Unemployment rate**
  The numerator is the unemployed and the denominator is the labor force. People outside the labor force therefore do not enter the denominator.
- **Employment rate**
  The employment rate divides the number employed by the working-age population in scope. It tells how much of that population works, regardless of whether people who do not work are looking for jobs.

Labor market indicators in Türkiye, 2019

Published count	Million people
Employed	28.080
Unemployed	4.469
Labor force	32.549
Outside the labor force	28.920
Population aged 15+ in scope	61.469

Counts are shown in millions of people.

- **Unemployment: 4.469 ÷ 32.549 = 13.7%**
  The 4.469 million unemployed are divided by the 32.549 million in the labor force. The result is 13.7% when rounded to one decimal place.
- **Participation: 32.549 ÷ 61.469 = 53.0%**
  For participation, the denominator changes to the 61.469 million people aged 15 and above included in the calculation. The displayed counts give 53.0% to one decimal, compared with the published rate of 52.9%.

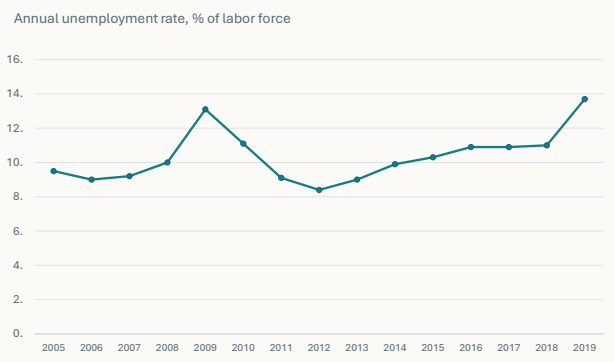
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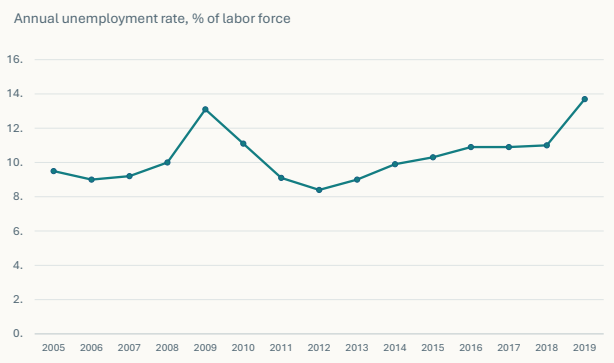
- **Employment: $28.080 \div 61.469 = 45.7\%$**
  The same population denominator is used, with 28.080 million employed in the numerator. The displayed counts give 45.7% to one decimal, compared with the published rate of 45.6%.

Unemployment over the business cycle



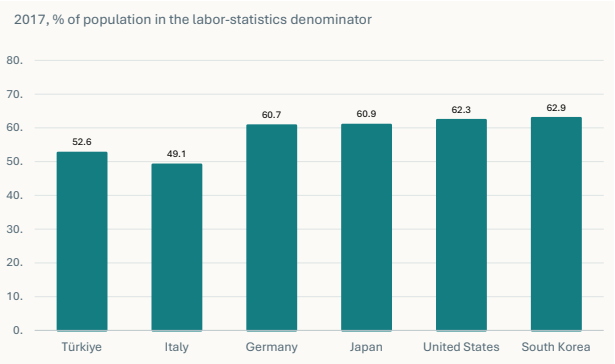
- **Unemployment rises sharply in 2009 and again by 2019.**
  Unemployment reached 13.1% in 2009 after the global crisis and 13.7% in 2019. In between, the rate fell to 8.4% in 2012 before rising again.
- **A decline in unemployment does not alone show how many adults work.**
  The rate can fall because job seekers find work or because people stop searching and leave the labor force. The employment rate uses the working-age population and helps distinguish those outcomes.

Unemployment over the business cycle



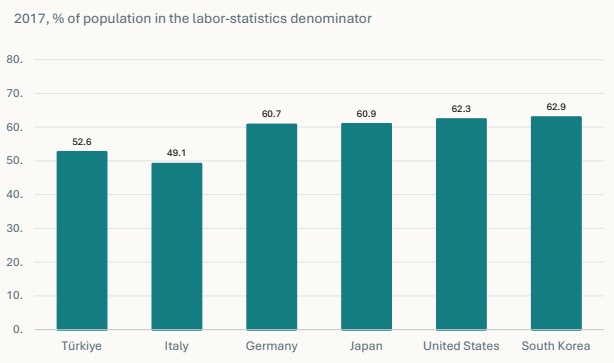
- **Participation and job quality add essential information.**
  Participation and unemployment should be interpreted together because movements into and out of the labor force can affect both rates. The employment definition covers different work arrangements, so a headcount alone cannot show job quality.

International comparison of labor force participation



- **Türkiye's reported participation rate is below several comparators.**
  In 2017, Türkiye's participation rate was 52.6%, below the rates in Germany, Japan, the United States and South Korea. Italy is below Türkiye, which is why the statement concerns several comparators rather than every country.
- **Care responsibilities, education and retirement patterns affect participation.**
  Household duties, study and retirement are among the reasons people may be outside the labor force. Differences in these circumstances can change participation even without a change in the unemployment rate.

International comparison of labor force participation



- **The unemployment rate captures only people already in the labor force.** It counts unemployed job seekers as a share of the employed and unemployed combined. Participation instead shows how large that combined group is relative to the working-age population in scope.

Inflation and purchasing power

- **Inflation (%)** $= \left(\dfrac{\text{CPI}_t}{\text{CPI}_{t-1}} - 1 \right) \times 100$

  The CPI measures the price of a representative basket relative to a base. Inflation is its percentage change between the selected dates, so the dates used determine whether the comparison is monthly or annual.

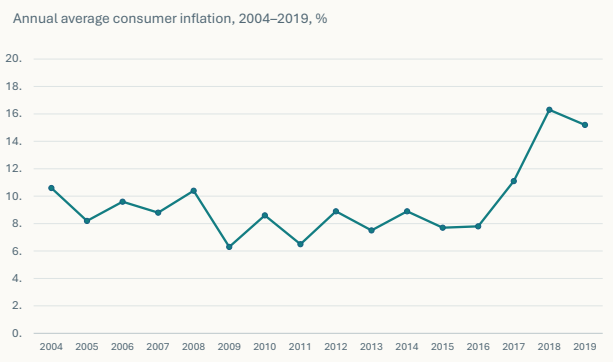
- **An index rise from 100 to 120 means 20% inflation.**

  An index increase from 100 to 107 implies 7% inflation. Applying the same method to 100 and 120 gives a 20% increase in the basket's cost.

Inflation and purchasing power

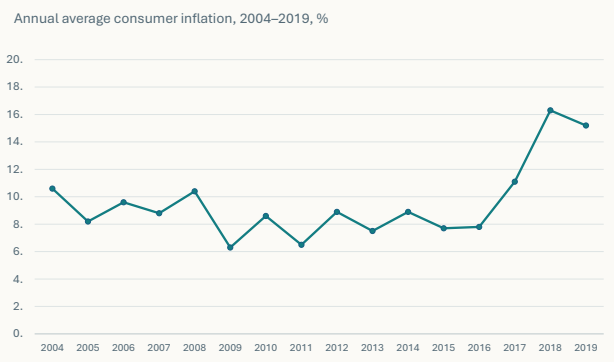
- **A nominal income rise of 10% with prices up 20% reduces real income by 8.3%.** Higher prices reduce purchasing power when income does not keep pace. Here, dividing the nominal income factor 1.10 by the price factor 1.20 gives 0.917 of initial purchasing power.
- **Lower inflation means slower price increases. It need not mean falling prices.** If the inflation rate falls but remains positive, the index continues to rise. A decline in the overall price level instead requires negative inflation, known as deflation.

Historical trends in consumer price inflation



- **Inflation fluctuates and rises into double digits in 2017–2019.**
  The historical CPI figure shows much lower inflation after the early 2000s than in the earlier high-inflation era. In the years highlighted here the reported annual averages are 11.1% in 2017, 16.3% in 2018 and 15.2% in 2019.
- **The 2019 rate is lower than in 2018, but average prices still rise.**
  The annual-average rate falls from 16.3% to 15.2%, a slowdown in the rate of increase. Because 15.2% is still positive, the 2019 annual-average price level remains above 2018's.

Historical trends in consumer price inflation



- **Annual-average and December-to-December inflation are different.**
  Inflation can be measured by comparing twelve-month averages or by comparing the current price index with a specific month, such as the previous December. The chart compares annual averages, whereas December-to-December uses only the two December index values.

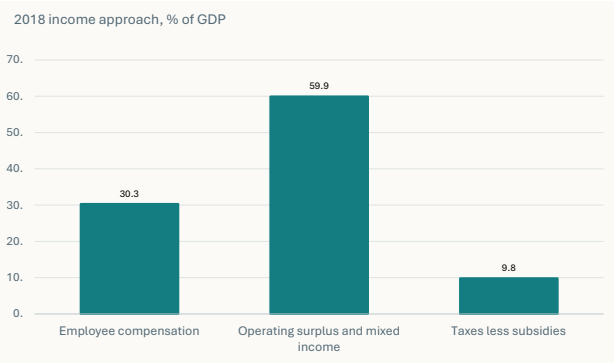
Economic effects of inflation and policy responses

- **Unanticipated inflation changes the real value of incomes and contracts.**
  Unexpected price increases reduce the purchasing power of payments fixed in nominal terms. When actual inflation differs from the rate anticipated in a contract, it can redistribute purchasing power between borrowers and lenders.
- **Uncertainty can shorten planning horizons and complicate investment.**
  Uncertain future prices increase risks for consumers, firms and investors. Inflation can favor short-term financial returns over investment in productive capital.

Economic effects of inflation and policy responses

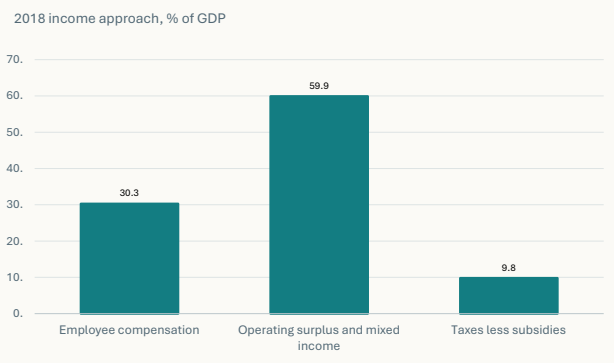
- **Demand pressures, costs, exchange rates and expectations can interact.**
  Inflation can be affected by demand, monetary conditions, oil costs, exchange rates and production constraints. Expectations can reinforce price increases when firms and workers adjust prices and contracts in anticipation.
- **Stabilization decisions involve timing, credibility and distributional effects.**
  Policy responses can emphasize monetary restraint or address structural bottlenecks. Inflation targets, realized inflation and distributional outcomes help assess whether the policy worked and who bore the adjustment.

Functional distribution of national income



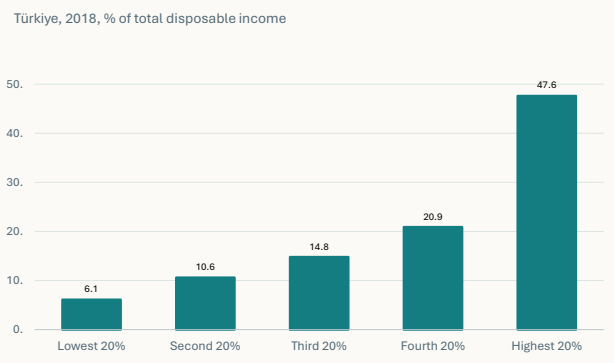
- **Functional distribution separates types of income.**
  "There are two different ways of measuring income distribution; one is personal and the other is functional." The 2018 income approach shows how much GDP takes the form of employee compensation, operating surplus and mixed income, or net production taxes.
- **Mixed income includes income where labor and capital returns are difficult to separate.** The 59.9% bar combines gross operating surplus and mixed income in one entry. A self-employed business owner can receive a single return that reflects both work and the use of capital, illustrating why the mixed component is difficult to split.

Functional distribution of national income



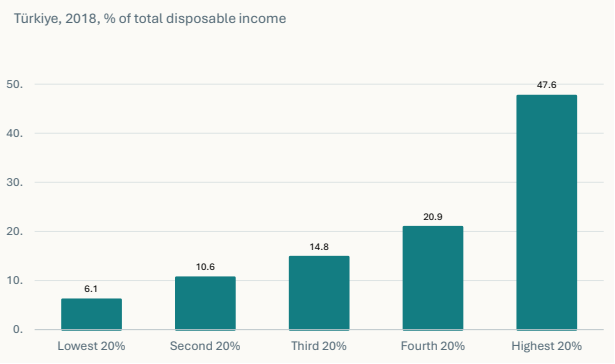
- **These categories are different from household income quintiles.**
  Functional distribution allocates income by type of return, such as employee compensation. Personal distribution groups households or people by income, including the combined income they receive from different sources.

Income shares by population quintile



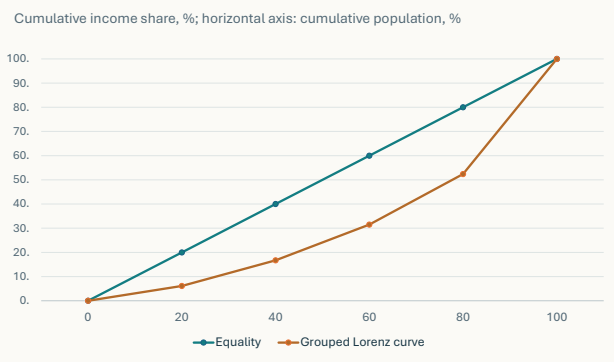
- **The highest-income fifth receives 47.6% of income.**
  A quintile contains one fifth of the ranked population. In 2018, the highest quintile received 47.6% of disposable income.
- **The lowest-income fifth receives 6.1%.**
  The lowest quintile contains the same population share as the highest quintile. Its income share was just 6.1%, showing why the overall average cannot describe the resources of each group.

Income shares by population quintile



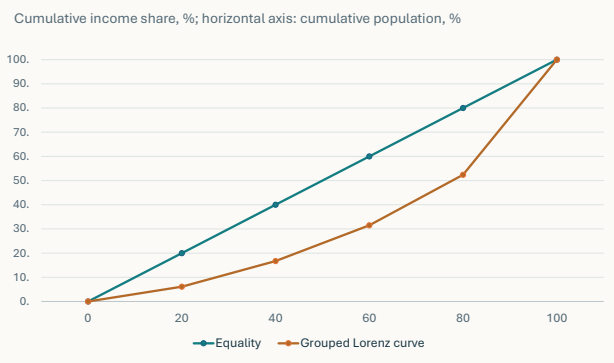
- **The ratio is about 7.8 to 1.**
  Dividing 47.6 by 6.1 gives about 7.8, the P80/P20 ratio reported for 2018. This ratio compares the two ends of the distribution and leaves the three middle quintiles out of the calculation.

The Lorenz curve and the Gini coefficient



- **The lowest 80% receive 52.4% of total income.**
  The Lorenz curve cumulates the income shares from the lowest to the highest group. Adding the first four quintiles gives 52.4%, the point on the curve at 80% of the population.
- **A curve farther below equality indicates greater inequality.**
  On the equality line, every cumulative share of people receives the same cumulative share of income. A larger area between that line and the Lorenz curve implies a higher Gini coefficient.

The Lorenz curve and the Gini coefficient



- **The 2018 Gini coefficient was 0.408.** The Gini is the area between the curve and equality divided by the total area below the equality line. The chart joins grouped quintile shares for illustration, while 0.408 is the published Gini coefficient.

Measures of poverty and income inequality

Concept	Question answered
Absolute poverty	Can people meet a specified minimum standard of living?
Relative poverty	How many people fall below a share of the median income?
Income inequality	How unevenly is income distributed?
Material deprivation	Which essential goods and activities can people not afford?

- **Absolute poverty**
  Absolute poverty is assessed against a minimum standard for meeting basic needs such as food, clothing and shelter. The poverty rate is the share of people below the specified threshold.
- **Relative poverty**
  A relative line is set as a fraction of the median disposable income for the reference population. Poverty comparisons can change when different thresholds or regional median incomes are used.

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- **Income inequality**
  Income inequality measures how income is shared across the entire distribution, using indicators such as the Gini or quintile shares. It does not by itself count the people below a poverty threshold.
- **Material deprivation**
  A deprivation approach asks whether people lack essentials such as adequate housing, sanitation or assets. Multidimensional poverty measures also consider education and health, which household income alone may not capture.

Development and public administration

Economic issue	Administrative connection
Low participation	Access to childcare, transport and skills
Regional productivity gaps	Infrastructure and local public services
Income insecurity	Design and delivery of social protection
Inflation exposure	Budget planning and public procurement
Environmental costs	Land use, energy efficiency and regulation

- **Low participation**
  Household responsibilities and education can limit participation in the labor force. Childcare, transport and training are classroom examples of services that could address barriers to work.
- **Regional productivity gaps**
  Productivity differs across sectors, and per capita income differs across regions. A public investment appraisal can use these patterns to ask whether infrastructure and local services improve access to productive work.

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- **Income insecurity**
  Pensions and unemployment payments are forms of social transfer. Poverty statistics help identify the groups that face income shortfalls. Administrators can use such evidence to assess coverage and whether support reaches households with greatest need.
- **Inflation exposure**
  Inflation raises the cost of the goods and services that households and public agencies buy. By tracking prices, procurement planners can distinguish price increases from changes in the quantities of services delivered.

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- **Environmental costs**
  Income alone does not capture environmental conditions or how natural resources are used. Land-use rules and energy-efficiency measures illustrate administrative decisions that should be judged on those outcomes as well as on output.

Türkiye within the global economy

International market	Main connection	Potential exposure
Goods and services	Trade and production networks	Foreign demand and input costs
Capital	Investment and external financing	Exchange-rate and refinancing risks
Labor	Migration, skills and remittances	Adjustment in jobs and public services

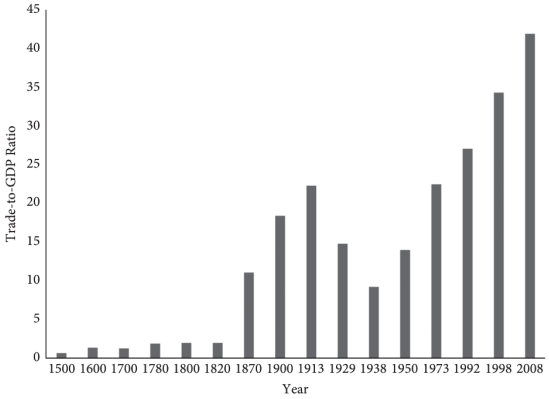
- **Goods and services**
  Trade openness links domestic production to imported inputs and foreign demand. A change in foreign demand or import costs can therefore reach domestic sectors through trade.
- **Capital**
  Part of the gap between Türkiye's national saving and investment was covered by external financing. Reliance on foreign financing can expose the economy to exchange-rate and funding pressures.

Türkiye within the global economy

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- **Labor**
  Migration and asylum can affect labor supply, the mix of skills and demand for services in host communities. This supports examining employment and service needs together, while remittances are a separate external income channel.

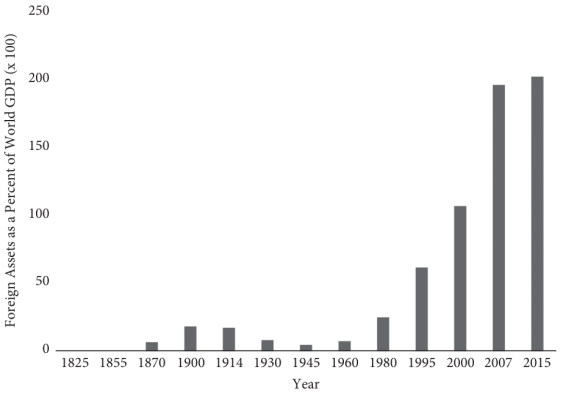
Long-run trends in world trade



World trade relative to GDP, 1500–2008.

- Trade integration rises over the long run.
- Wars and the interwar period interrupt that expansion.
- Technology and policy both shape trading costs.

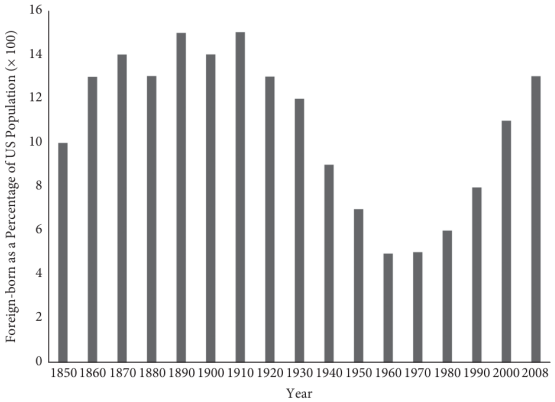
Long-run trends in financial integration



Foreign assets relative to world GDP, 1820–2015.

- Foreign assets are a stock measured relative to annual GDP.
- The ratio falls during the early twentieth-century disruption.
- It rises strongly in the later period shown.

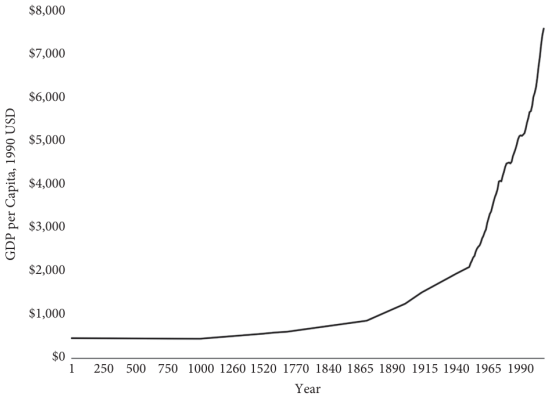
International migration: the United States



Foreign-born share of the US population, 1850–2008.

- The US foreign-born share falls in the mid-twentieth century and later recovers.
- This is a population stock, not annual immigration.
- Labor integration follows a different path from goods or finance.

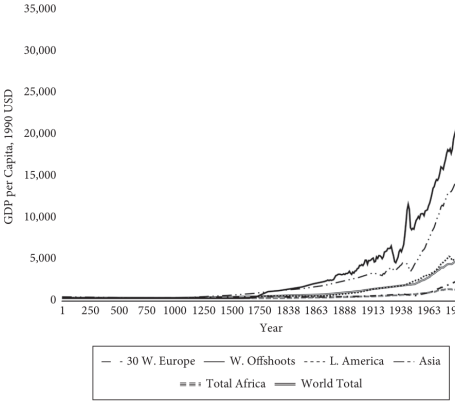
Long-run trends in world GDP per capita



World GDP per capita, 1–2008 CE.

- Sustained growth in income per person emerges late in this long history.
- Technology and institutional change help explain the transition.
- The world average conceals major differences across societies.

Regional divergence in GDP per capita



Real GDP per capita by region, 1–2008 CE.

- Western Europe and Western offshoots pull ahead early.
- Other regions follow different growth paths.
- Trade is one influence among technology, institutions and historical conditions.

Gains and adjustment costs of economic integration

Potential gain	Potential cost or condition
Cheaper inputs and more product choice	Some domestic producers face stronger competition
Larger markets and specialization	Workers may need new skills or locations
Capital and technology transfer	Foreign-currency or refinancing exposure
New opportunities for migrants	Adjustment pressures and public-service needs

Economic effects of an imported energy price shock

- **What happens when the foreign-currency price of energy rises?**
  Much of Türkiye's demand for oil and gas is met by imports. Holding volumes and the exchange rate fixed, a higher foreign price increases the cost of those imports.
- **Trace effects on production costs, household budgets and external payments.**
  Energy-using firms face higher input costs, which may be passed on in consumer prices. Higher spending on imported energy also puts pressure on external payments unless quantities or other transactions adjust.

Economic effects of an imported energy price shock

- **Compare temporary household support and investment in efficiency.**
  Temporary support can protect some households' purchasing power during the shock. Efficiency investment takes time to implement but can reduce energy demand per unit of output and thereby lower future exposure.
- **Who benefits, who pays, and what evidence would you need?**
  Household spending patterns and dependence on imported energy help identify who is most exposed to the shock. Compare support costs and household coverage with the expected energy savings and implementation costs of investment.

Summary of economic indicators

- **Measurement comes first: units, dates and definitions change the interpretation.**
  Real output, current-dollar values and purchasing power are distinct measures. CPI changes depend on the comparison period, and labor-market rates depend on the denominator used.
- **Production, employment and external finance are closely connected.**
  Comparing sectoral employment and output helps locate differences in measured productivity. The saving gap and the composition of imports show how activity and investment also depend on external resources.

Summary of economic indicators

- **Growth and integration can raise income while distributing gains unevenly.**
  Rising per capita income does not establish that living conditions improve for everyone. Income shares by quintile and poverty measures help assess how the gains from economic change are distributed.
- **Exit question: which two additional indicators would you request after hearing "GDP grew"?**
  One useful pair is real GDP per capita growth and the employment rate, since they show changes in output per person and the share of adults working. Depending on the question, inflation or the lowest quintile's share can add evidence about purchasing power and distribution.